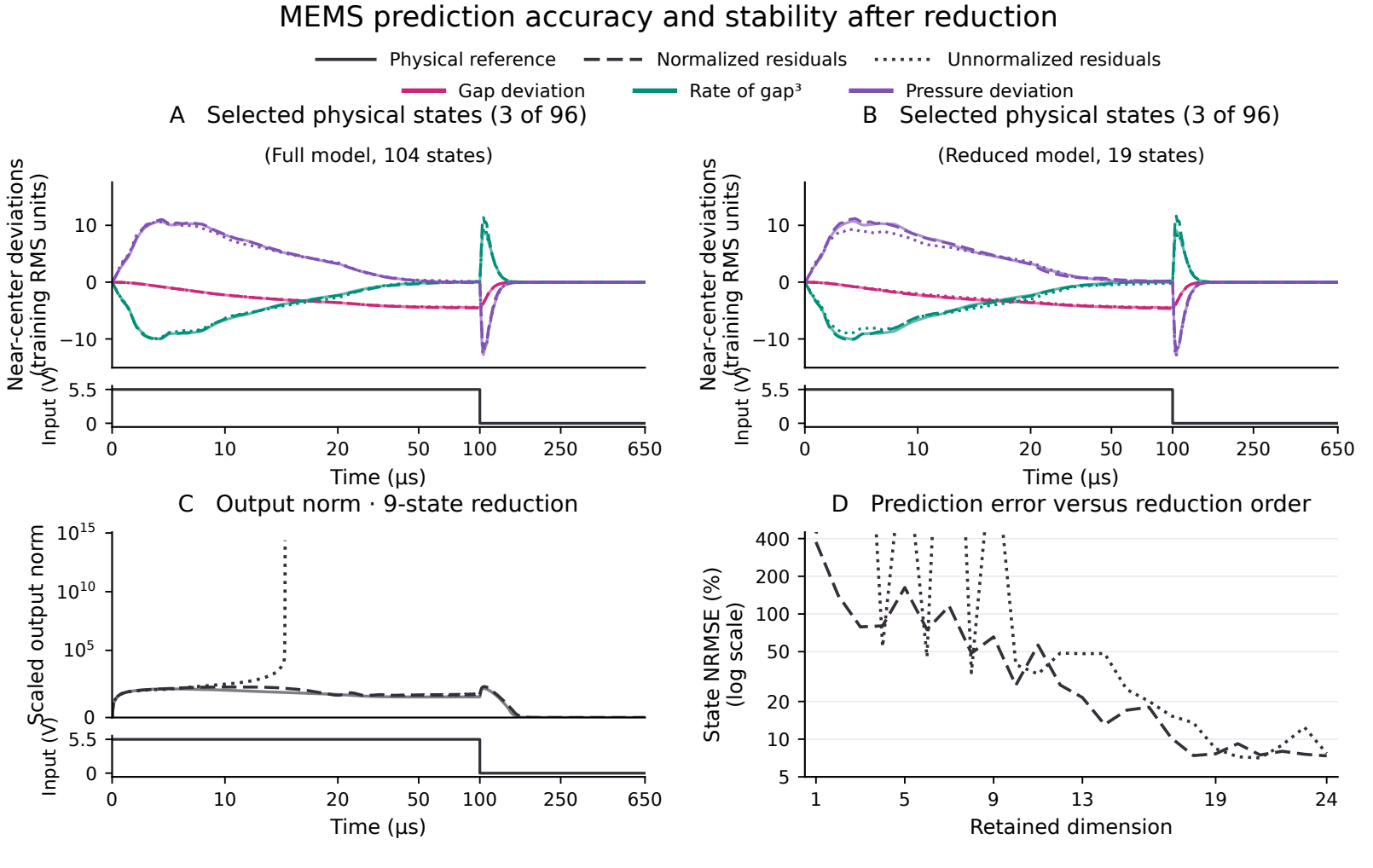
MEMS prediction accuracy and stability after reduction
Physical reference
Normalized residuals
Unnormalized residuals
Gap deviation
Rate of gap³
Pressure deviation
A Selected physical states (3 of 96)
(Full model, 104 states)
B Selected physical states (3 of 96)
(Reduced model, 19 states)
Near-center deviations (training RMS units)
Input (V)
Time (μs)
C Output norm · 9-state reduction
Scaled output norm
D Prediction error versus reduction order
State NRMSE (%) (log scale)
Retained dimension